

**Project abstract (up to 500 words). Include a brief description of the unmet need and proposed medical device. (By 7/19 11:59 PM)**

**Link To Letter of Intent:** [Here](#)

Heart failure is a leading cause of mortality in the United States, often resulting from various disorders that impair cardiac function. It presents a significant and costly health challenge, with many patients requiring advanced therapies such as a left ventricular assist device (LVAD). An LVAD functions as a mechanical pump, with an inflow conduit inserted into the left ventricle and an outflow grafted to the ascending aorta. Despite advancements in LVAD design, complications remain prevalent. Patients with an LVAD experience a high rate of hospital readmission, the primary issue with current LVAD systems being the high risk of sepsis or driveline infections [1], often occurring more than 30 days post-implantation, likely due to patient contamination rather than clinical error.

LVADs are typically powered via rechargeable batteries that last around 14 hours. Our proposed improvement on the LVAD system is a *polymagnetically-coupled inductive charging system* that powers the device through the patient's skin. Wireless charging coils generate an alternating electromagnetic field that transfers energy via induction to another nearby subcutaneous coil. The implanted LVAD device has a small battery and controller circuit, as well as a receiver coil positioned beneath the skin. This coil is magnetically connected to a transmitter coil and controller/battery pack module embedded within a wearable vest outside of the patient's body. Misaligning the coils causes a decrease in the energy transmitted and an increase in temperature. To address these problems, we plan to surround the charging coils in smart magnets, which exclusively attract and connect to the corresponding smart magnet, to avoid misalignment of the coil. These magnets would consist of specifically sequenced north south patterns that only attract the matching magnet sequence. We will also use dynamic power delivery and adaptive tuning techniques to regulate the temperature of the battery to not exceed body temperature and cause discomfort [2].

With a fully implantable system, patients will experience fewer complications while using LVAD. Two medical device companies, Abbott and Medtronic, are currently in the process of developing fully implantable LVAD systems [3] but lack polymagnet technology [2]. This new technology will more securely align the transmitter and receiver coils without risk of attaching to everyday ferromagnetic objects. With increased coil alignment and stabilization, a fully

implanted LVAD system will reduce risk of driveline infection and improve LVAD patients' quality of life.

- [1] Yarboro, Leora T., et al. "Technique for Minimizing and Treating Driveline Infections." *Annals of Cardiothoracic Surgery*, vol. 3, no. 6, 1 Nov. 2014, pp. 557–562, [www.ncbi.nlm.nih.gov/pmc/articles/PMC4250551/](http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4250551/), 10.3978/j.issn.2225-319X.2014.09.08.
  
- [2] Stein, Aaron, et al. "CARD14: Wireless Transcutaneous Energy Transfer and Control System for Powering a Fully-Implanted Left-Ventricular Assist System." *Asaio Journal*, vol. 69, no. Supplement 2, 1 June 2023, pp. 64–64, <https://doi.org/10.1097/01.mat.0000943580.83459.85>.
  
- [3] "Abbott Wins Breakthrough Tag for Fully Implantable LVAD System, Following Medtronic." *MedTech Dive*, [www.medtechdive.com/news/abbott-wins-breakthrough-tag-for-fully-implantable-lvad-system-following-m/571670/](https://www.medtechdive.com/news/abbott-wins-breakthrough-tag-for-fully-implantable-lvad-system-following-m/571670/).